BinWatch

Which rubbish bin in The Hague needs a crew first?

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AI for Good, Hackathon 7 · SDG 15 Life on Land · Built with Roboflow

Bags next to the bin, gulls get there first

What happens

People leave rubbish bags next to the underground containers. Gulls tear them open and the rubbish spreads into streets, parks and canals.

What the city says

Don't put bags next to the containers. The municipality even gives out free, extra strong yellow bags, because gulls peck normal ones open.

The blind spot

The containers have sensors that report when they are full. But a sensor only sees what is inside, not the bags lying next to it.

11.2 million kg

illegally dumped bulky waste in The Hague in 2025, 0.6 million kg more than in 2024. Misplaced bags went up too.

37%

of container locations in the Netherlands had rubbish left next to them in 2024.

Sources: Gemeente Den Haag, Meeuwenoverlast beperken · Waste Action Plan progress report 2025 (The Hague Online, 24-04-2026) · NVRD (NL Times, 03-04-2026)

For the route planners, not for fines

Who it is for

The municipal team that plans where the waste crews go. Residents see bags next to a bin and send in a photo. On a busy day there are too many reports to go everywhere at once.

- BinWatch puts the reported bins in order: **Clear first**, then **Clear on next round**. **Clean** can wait.

Who it is not for

- Enforcement officers (handhavers) who want to give fines
- Anyone who wants to watch a street or a resident
- Counting gull damage: the model sees rubbish, not who or what caused it

From photos to a clear-up order

1

Photos

One or more photos of bins, uploaded or taken with the phone camera.

2

Roboflow

Our model boxes each object: bag, torn_bag, loose_litter, dumped_item. A slider hides unsure boxes.

3

Priority rule

The app turns the labels it got back into one priority.

4

Result

Each photo with boxes, a count per class and the priority. With several photos: a Clear-up order.

- **Clear first:** a torn bag or loose litter. The rubbish is already spreading.
- **Clear on next round:** closed bags or dumped items next to the bin, nothing torn yet.
- **Clean:** nothing found. This bin can wait.

Why not something simpler? Fill sensors can't see next to the bin, and driving the same route every day wastes time on bins that are already clean.

Our own data, filmed in The Hague

7
videos filmed by us

3–4 / sec
frames taken from the videos

300
labelled images

4
classes

How we made it

- Filmed in October, in daylight only
- Removed blurry frames and frames that looked almost the same
- Clean bins kept as "null" images, so the model also learns what "nothing to do" looks like

Kept honest

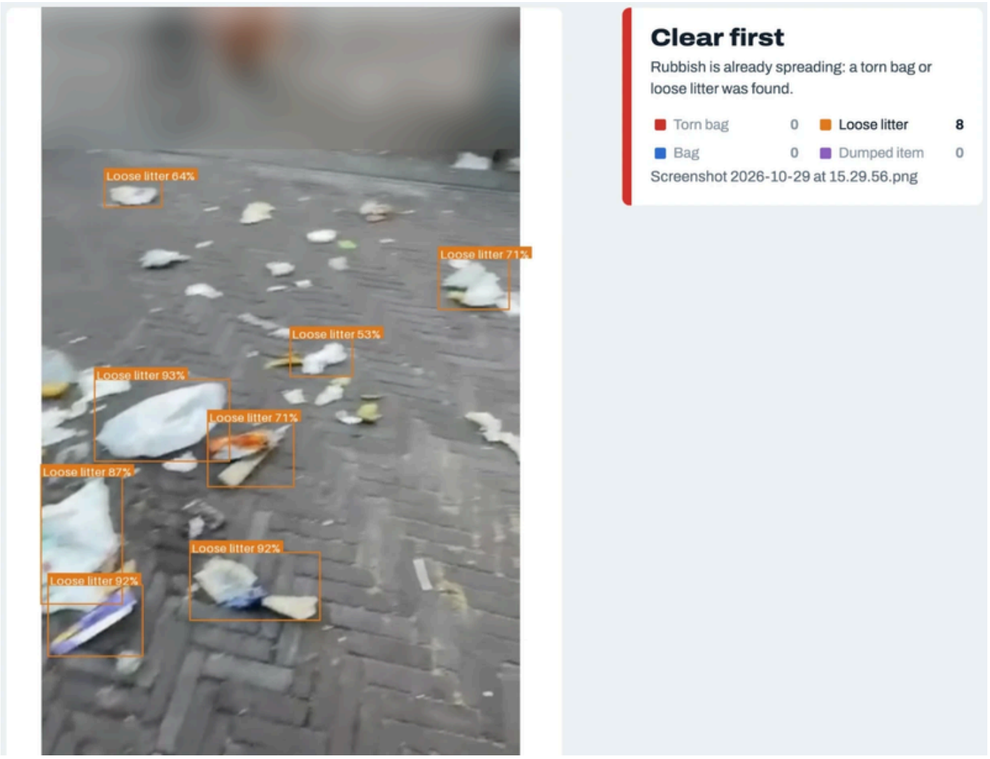
- Removed 21 frames with a readable number plate
- Split: 211 train / 60 valid / 29 test
- Test images come from the same videos as training, so real-world scores will be lower

Results and live demo

Class	Found	Precision	Recall
All classes		**88.5%**	**86.4%**
bag	12 / 13	92%	92%
torn_bag	9 / 9	100%	100%
loose_litter	6 / 9	75%	67%
dumped_item	13 / 15	87%	87%

mAP@50, all classes: 89.6%
Hardest class: loose_litter, 6 of 9 found and 2 false alarms. Small scattered pieces are hard to see.

Roboflow test set (29 images). Per-class numbers are counted from Roboflow's confusion matrix. The test images come from the same videos as training, so real-world scores will be lower.



Our test: 8 × loose litter, Clear first. At 0.40 it found the bigger pieces but missed many small ones.

Live demo: binwatch.streamlit.app

What can go wrong

Limits

- 300 images from a few streets
- Test images come from the same videos, so real scores will be lower
- Only autumn, but gulls are worst in spring and summer
- WhatsApp made the videos less sharp

Failure modes

- It sees damage, not the cause: a bag torn by a rat, cat or person looks the same
- It misses small litter, and it has never seen night photos
- A wrong "Clean" is the worst mistake: a dirty bin waits

Who takes the photos

- Residents report bins, but some areas report more than others
- Photos can show neighbours, house numbers, number plates
- Someone could use it against a neighbour

What we do about it

Done: 21 frames with number plates removed, the app says it's only a hint for route planning, never proof for fines. **Next:** test on bins we never filmed, only order reported bins (normal route stays), blur faces and plates, delete photos after clearing.

What we learned and what's next

What we learned

- Who takes the photo matters as much as the model
- The model is good at big pieces but misses small litter
- A high score in Roboflow doesn't mean it works on a new street

Next steps

- Test on photos of bins we never filmed
- Film more torn bags, and film in spring and summer
- Blur faces, number plates and house numbers before a photo is saved
- Film at night too

Thank you. Questions?